WILBERT ANDREW YONATHAN

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CAREER OBJECTIVE

Final-year Artificial Intelligence in Engineering student at Xiamen University Malaysia with experience in computer vision, reinforcement learning and efficient AI. Seeking an AI Engineering or Computer Vision internship to develop solutions through research and problem-solving while advancing technical skills.

EDUCATION

Xiamen University Malaysia Sep 2023 – Sep 2027 (Expected)
Bachelor of Engineering in Artificial Intelligence | **CGPA: 3.77/4.00**

TECHNICAL SKILLS

Programming: Python, C, C++
AI/ML: Computer Vision, Deep Learning, Object Detection, Anomaly Detection, Reinforcement Learning, RAG, LLMs
Frameworks: PyTorch, TensorFlow, YOLO, OpenCV, scikit-learn, LangChain, LangGraph, FastAPI, React, Gradio, Unity ML-Agents
Tools: Git, Docker, Supabase, Qwen, Hugging Face ZeroGPU, Render, Streamlit
Languages: English, Mandarin, Indonesian, Malay, Hokkien

PROJECT EXPERIENCE

PaddyLiteX: Paddy Disease Detection Apr 2026 – Present
- Developed a DenseLiteX-SPPF-C2PSA YOLOv8n detector for nine paddy classes, achieving 81.67% mAP@0.5 (+0.88 percentage points) with 4.24% fewer parameters than YOLOv8n.
- Compared with the initial YOLOv8n-DenseNet121 proposal, reduced parameters by 68.01% and GFLOPs by 72.59%; conducted ablation studies and deployed a three-model comparison demo.

DefectRAG: Visual Anomaly Inspection and RAG Aug 2026 – Oct 2026
- Developed a DINOv2 visual anomaly detector across 12 VisA products, achieving 95.13% mean image AUROC in retrospective evaluation with anomaly heatmaps.
- Integrated LangGraph, BGE and Qwen for cited RAG explanations in a React dashboard with Supabase review workflows and Hugging Face ZeroGPU deployment.

MarineLite: Marine Organism Detection Jun 2026 – Jul 2026
- Integrated GhostConv and SimSPPF into DU-MobileYOLO, improving mAP@0.5 from 74.23% to 75.36% while reducing parameters by 7.02%.
- Conducted ablation studies to assess accuracy–efficiency trade-offs and deployed a web demo with detection overlays and model comparisons.

dPickleball: Reinforcement Learning Agent May 2026 – Jul 2026
- Trained PPO agents in Unity ML-Agents using a six-level curriculum, reward shaping and league self-play to improve autonomous paddle control.
- Contributed policies and tournament evaluation to a team submission that placed 2nd among 13 teams.

WORK EXPERIENCE

Student Registration & Operations Assistant, Xiamen University Malaysia Apr 2025
- Supported registration logistics and directed students as part of a team serving 800+ o students.
- Organized university assets and materials for event operations and proper storage.

LEADERSHIP AND ACTIVITIES

Head of Photography & Videography, Love in Action Feb – Mar 2026
Coordinated media coverage, workflows and edited outreach materials.
Head of General Affairs, XMUM AI Club Apr – Oct 2025
Coordinated event logistics and external partners for a national-level AI event.

AWARDS

- Dean's List: Awarded for three consecutive semesters.
- XMUM Merit Scholarship: Awarded for the four-year undergraduate programme (2023–2027).

REFERENCES

Shaidah Binti Jusoh, Assistant Professor | Thesis Supervisor
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Goh Sim Kuan, Assistant Professor | Internship Coordinator
simkuan.goh@xmu.edu.my | 012-7660576